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A novel proliferative candidate genes panel for idiopathic pulmonary fibrosis: insights from integrated bulk and single-cell RNA sequencing

Short Title: Proliferative candidate genes panel for IPF

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Abstract

Background： Idiopathic pulmonary fibrosis (IPF) remains a fatal interstitial lung disease with limited diagnostic specificity and therapeutic options. This study integrates bulk and single-cell RNA sequencing (RNA-seq) to identify novel biomarkers and elucidate molecular mechanisms underlying IPF pathogenesis. **Methods：** We prospectively enrolled 14 treatment-naive IPF patients and 6 controls. Bulk RNA-seq was performed on bronchoalveolar lavage fluid (BALF), while single-cell RNA-seq analyzed lung tissues from 4 IPF patients and 3 controls. Differentially expressed genes (DEGs) were identified (|log2FC| >1, FDR <0.05), followed by functional enrichment, protein-protein interaction (PPI) network analysis, and cell-type-specific expression profiling. **Results：** **1.** DEG Identification: Bulk RNA-seq revealed 108 DEGs (24 upregulated, 84 downregulated). KEGG enrichment analysis of DEGs revealed that upregulated genes were mainly enriched in inflammation and immune pathways (such as NF-κB signaling pathway, Fc epsilon RI signaling pathway, B cell receptor signaling pathway, phagosome, Fc gamma R-mediated phagocytosis), pyrimidine metabolism, cell cycle, and

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PI3K-Akt signaling pathway. 2. PPI Network: Module analysis identified a proliferative gene module 1 (NUF2, CEP55, ANLN, TTK, TK1, MYBL2, CCNA2, RRM2, CDT1) linked to cell division and cycle regulation. 3. Single-Cell Insights: scRNA-seq of 30,477 cells delineated 11 populations. Module 1 genes exhibited predominant expression in proliferating cells, Module 1 signature score of proliferating cells was significantly higher in IPF than in control group. 4. Pathogenic Links: Key genes (e.g., CEP55, TTK) were associated with PI3K/AKT signaling, epithelial-mesenchymal transition (EMT), and anti-apoptotic pathways, mirroring oncogenic mechanisms.

Conclusion： This multi-omics approach uncovers a proliferation-centric gene module in IPF, revealing shared molecular pathways with tumorigenesis. Our findings highlight novel diagnostic biomarkers and suggest repurposing cell cycle inhibitors as potential therapies. Future studies should validate these targets in preclinical models to advance precision medicine for IPF.

Keywords: Idiopathic pulmonary fibrosis; bulk RNA sequencing, single-cell RNA sequencing RNA sequencing; Biomarkers; Cell cycle; Proliferating cells.

1 Introduction

Idiopathic pulmonary fibrosis (IPF) is a chronic, progressive, and fatal interstitial lung disease (ILD) with a median survival of 3–5 years following diagnosis [1]. Characterized by aberrant lung fibrogenesis and histopathological patterns consistent with usual interstitial pneumonia (UIP), IPF manifests with debilitating symptoms—including persistent cough and exertional dyspnea—that profoundly impair health-related quality of life. Diagnosis requires a two-pronged approach: (1) multidisciplinary exclusion of known ILD etiologies through clinical-radiological-pathological correlation, and (2) confirmation of UIP on high-resolution computed tomography (HRCT) or surgical lung biopsy (SLB) / transbronchial lung cryobiopsy (TBLC) . TBLC was regarded as an acceptable alternative to SLB for making a histopathological diagnosis in patients with ILD of undetermined type in medical centers with experience performing and interpreting TBLC (conditional recommendation) [1].

However, diagnostic accuracy remains constrained by two critical limitations:

1. **Radiological ambiguity**: HRCT demonstrates suboptimal specificity in differentiating IPF from other fibrotic ILDs (e.g., chronic hypersensitivity pneumonitis [cHP], connective tissue disease-associated ILD [CTD-ILD]), which may exhibit radiologically overlapping UIP-like patterns [2,3].
2. **Procedural risks**: Although SLB is diagnostic gold standard, its invasiveness precludes use in patients with advanced disease or comorbidities due to high morbidity

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(e.g., pneumothorax, acute exacerbation) and mortality rates (3–6%) [1]. Existing biomarkers (e.g., Krebs von den Lungen-6 [KL-6], surfactant protein-A/-D [SP-A/D]) exhibit limited specificity [4,5]. These gaps underscore the urgent need for novel biomarkers with high diagnostic accuracy.

Pathobiological insights

IPF has undergone a paradigm shift from initial attributions to chronic inflammatory processes [6] toward recognition of the significant role played by oxidative stress [7]. Contemporary models posit that repetitive microinjuries to the alveolar epithelium induce cellular dysfunction, triggering a self-perpetuating cascade of fibroblast activation, proliferation, and extracellular matrix (ECM) overproduction through dysregulated pro-fibrotic signaling [8,9]. This maladaptive repair process culminates in irreversible parenchymal scarring and progressive loss of pulmonary function. [10,11] Despite the efficacy of antifibrotic therapies (pirfenidone, nintedanib) in slowing forced vital capacity (FVC) decline, IPF remains incurable [1]. These limitations highlight the imperative to elucidate novel molecular mechanisms and therapeutic targets.

Rationale for the present study

Bronchoalveolar lavage fluid (BALF), a minimally invasive biospecimen, offers a unique window into the alveolar microenvironment. When combined with bulk and single-cell RNA sequencing (RNA-seq), BALF enables high-resolution mapping of disease-specific transcriptomic signatures. While bulk RNA-seq identifies population-level expression trends, scRNA-seq resolves cellular heterogeneity, thereby overcoming the limitations of traditional tissue-based approaches.

Study hypothesis: Integrative analysis of bulk and scRNA-seq data from BALF and lung tissue will identify novel IPF biomarkers with diagnostic potential.

2. Methods (Technical Route Outlined in Figure 1)

2.1 Study Cohort

We prospectively enrolled 14 treatment- naive IPF patients (2023–2024) and 6 age-/sex-matched controls with benign pulmonary nodules from Kunming First People's Hospital. Inclusion criteria:

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- IPF diagnosis based on ATS/ERS/JRS/ALAT guidelines (an Update) [1].
- Exclusion of alternative ILDs via multidisciplinary discussion.

Data collection:

- **Demographics**: Age, sex, smoking history (pack-years).
- **Radiological**: HRCT pattern: UIP, Probable UIP, Indeterminate, Alternative diagnosis.
- **Pulmonary function**: FVC, FEV_1, DLCO (predicted %), TLC.
- **Laboratory**: BALF differential counts, serum KL-6, C-reactive protein [CRP], lactate dehydrogenase [LDH], the autoantibody panel((ENA profile, ANCA, and rheumatoid-associated antibodies).
- **Ethics**: Approved by Kunming First People's Hospital IRB (No. YLS2022-82). All participants provided written informed consent.

2.2 Bulk RNA-Seq and Bioinformatics

- **Sample processing**: BALF cells from IPF (n=14) and controls (n=6) underwent poly(A)+ RNA-seq (Illumina NovaSeq 6000, 150 bp paired-end).
- **Differential expression**: Differentially Expressed Genes (DEGs) were were screened using False Discovery Rate (FDR) and log2FC, with FDR < 0.05 and |log2FC| > 1. Genes with FC (i.e., fold change) greater than 2 were defined as DEGs, and genes with Log2FC>1, i.e., FC greater than 2, were considered up-regulated, and those with Log2FC＜-1, i.e., FC less than 0.5, were considered down-regulated.
- **Functional enrichment**: Kyoto Encyclopedia of Genes and Genomes (KEGG) and Gene Ontology (GO) analysis (clusterProfiler, R).

2.3 PPI Network and Module Analysis

- **Network construction**: STRING database (confidence score >0.7).
- **Module identification**: MCODE plug-in (Cytoscape) with parameter settings: degree cutoff=2, node score cutoff=0.2, k-core=2, max. Depth=100.

2.4 scRNA-seq and Bioinformatic Analysis

- **Tissue acquisition**: Transbronchial lung biopsies (TBLB) from 4 treatment-naïve IPF patients (2023–2024) and 3 controls with benign pulmonary nodules
- **Sample Preparation**:
    - Single-cell suspension (1×10^5 cells/mL) in PBS (HyClone).
    - Single-cell capture using GEXSCOPE® microfluidic chip.
    - Addition of beads with unique cell barcodes to microwells (1 bead/well).

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- **mRNA Capture & Library Construction**:
  - Post-lysis, mRNA is captured by beads tagged with cell barcodes and UMIs.
  - mRNA reverse-transcription to cDNA, followed by amplification.
  - Library construction using GEXSCOPE® Single-Cell RNA Library Kit.
- **Sequencing**:
  - Library diluted to 4 nM.
  - Sequenced on Illumina HiSeq X (150 bp paired-end).
- **Data Analysis**:
  - **Quantification**:
    - Group reads/UMIs/genes by cell barcode; count UMIs per gene/cell.
  - **Clustering & Cell Type Identification**:
    - Use Seurat (R package, v.3.0.1) for analysis.
    - Import expression matrix via read.table; cluster cells using FindClusters (resolution = 0.6).

3. Results

3.1 Characteristics of Enrolled Subjects

Compared with the control group, the IPF group showed no significant differences in age, gender composition, or smoking history. Among the four parameters of lung function, the percentage of measured to predicted value of (diffusing capacity for carbon monoxide, DLCO) (DLCO%) and (total Lung Capacity, TLC) (TLC%) were significantly lower in the IPF group than in the control group. However, there were no significant differences in the percentage of measured to predicted value of (forced vital capacity, FVC) (FVC%) and (forced expiratory volume in the first second, FEV1) (FEV1%) between the IPF group and the control group. The total number of white blood cells and the absolute value of neutrophils in peripheral blood were significantly higher in the IPF group than in the control group, while the percentage of lymphocytes was significantly lower in the IPF group. Additionally, the levels of cytokeratin 19 fragment and LDH were significantly higher in the IPF group compared to the control group (Table 1).

3.2Statistics of DEGs between Groups

Differential expression analysis between the IPF group and the control group revealed 108 DEGs with a fold change >2, including 24 upregulated and 84 downregulated genes (as shown in Table 2). Among the 84 downregulated genes, 4 were unknown genes. The analysis of DEG expression patterns between groups is shown in Figure 2.

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3.3 Results of Pathway and Functional Enrichment Analysis

GO enrichment analysis of DEGs showed (as in Figure 3) that upregulated DEGs were mainly enriched in biological processes such as cell cycle, organelle organization, mitotic cell cycle process, cell division, protein complex biogenesis, macromolecular complex subunit organization, cellular component assembly, and small molecule metabolic process. Downregulated DEGs were mainly involved in small molecule metabolic process, cellular component assembly, macromolecular complex subunit organization, lipid metabolic process, and cell cycle.

The results of KEGG enrichment analysis of DEGs (as shown in Figure 4) revealed that upregulated genes were mainly enriched in inflammation and immune pathways (such as NF-κB signaling pathway, Fc epsilon RI signaling pathway, B cell receptor signaling pathway, phagosome, Fc gamma R-mediated phagocytosis, intestinal immune network for IgA production, natural killer cell-mediated cytotoxicity), pyrimidine metabolism, cell cycle, and PI3K-Akt signaling pathway. Among them, inflammation and immune response seemed to be the most enriched pathways. The most significantly enriched pathway for downregulated DEGs was the cell adhesion molecules (CAMs) pathway, followed by pathways related to immune-related diseases (IgA production intestinal immune network, allograft rejection, systemic lupus erythematosus, autoimmune thyroid disease, and inflammatory bowel disease). These findings highlight the complexity of the immune microenvironment in IPF, which is likely associated with immune dysregulation. This provides new insights for developing therapeutic strategies targeting the immune microenvironment. For example, IgA plays a key role in maintaining immune tolerance in mucosal immunity, including in the respiratory and intestinal tracts. Dysregulation of its production network may indicate impaired mucosal immune tolerance in the lungs of IPF patients, which could be one of the early events in disease pathogenesis. Studies on IgA suggest that it may promote human lung inflammation and fibrosis by enhancing the production of inflammatory or fibrogenic cytokines as well as extracellular matrix, inducing fibroblast differentiation into myofibroblasts, and promoting the proliferation of human lung fibroblasts[12]. This conclusion partially diverges from the findings of our study, and further research is needed to explore the role of these immune-related mechanisms in the pathogenesis of IPF.

3.3 PPI Network Construction and Module Analysis

The 104 DEGs were input into the STRING database to obtain the PPI network diagram. After organizing the PPI network data, it was imported into Cytotype, and the MCODE plugin was used for further clustering and module analysis (parameter settings: degree cutoff=2, node score cutoff=0.2, k-core=2, max. Depth=100). The module with the largest calculated Score

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value was taken as the final functional module, and three important modules were discovered (as shown in Figure 5), resulting in 17 MCODE genes occupying the most important positions in the PPI network. Among these, nine MCODE genes—NUF2, CEP55, ANLN, TTK, TK1, MYBL2, CCNA2, RRM2, and CDT1—were identified as upregulated (highlighted in red) and constituted Module 1, mainly involved in cell division, cell cycle, and their regulation (Table 3). Eight MCODE genes were downregulated (depicted in blue), with four genes (TBC1D8B, KPNA5, KIAA1328, and RPAP2) forming Module 2, and the remaining four genes (HLA-DRB1, HLA-DMA, HLA-DPA1, and HLA-DRA) constituting Module 3

3.4 Relationship between IPF Severity and Module 1 Genes

Correlation analysis was performed between the gene expression levels of NUF2, CEP55, ANLN, TTK, TK1, MYBL2, CCNA2, RRM2, and CDT1 in bronchoalveolar lavage fluid (BALF) and the pulmonary function parameters DLCO%, TLC%, FVC%, and FEV1%. This analysis revealed that both NUF2 and TTK expression were negatively correlated with TLC% (Figure 6). Patients were dichotomized into two groups (severe group vs. mild group) based on a threshold of 60% for each of the four pulmonary function parameters. Comparing the gene expression levels of the aforementioned genes in BALF between these groups revealed that NUF2(0.94±.513 VS 0.255±0.205) and ANLN(1.347±0.459 VS 0.51±0.370) expression were significantly higher in the severe group compared to the mild group.

3.5 Single-Cell RNA Sequencing and Analysis of Target Gene Expression Profiles Across Diverse Cell Types in Lung Tissue

We performed high-resolution single-cell RNA sequencing analysis on 30,477 cells isolated from lung tissues of 4 treatment- naive IPF patients and 3 controls (demographics summarized in Table 4). Unsupervised clustering analysis identified 11 major cell populations (Figure 7): mononuclear macrophages "MPs", ciliated cells "CiliatedCells", secretory cells "SecretoryCells", basal cells "BasalCells", alveolar epithelial cells "AlveolarEpithelialCells", endothelial cells "ECs", proliferating cells "Proliferatingcells", plasma cells "PlasmaCells", T cells "TCells", mast cells "MastCells", and stromal cells "StromalCells". Each cell cluster was labeled and identified based on known cell markers (Table 5). Each cell cluster could be further subdivided into subpopulations, and here we mainly highlight the subpopulation data within four pivotal cell types: MPs, ProliferatingCells, StromalCells, and AlveolarEpithelialCells.

As seen in Figure 8, among the upregulated DEGs in IPF compared to the control group, the nine genes (NUF2, CEP55, ANLN, TTK, TK1, MYBL2, CCNA2, RRM2, and CDT1) constituting Module 1 all showed the highest expression levels in "Proliferatingcells"(Figure

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8A). The expression profiles of these genes across proliferating cell subclusters are shown in Figure 8B. Next, we computed Module 1 genes signature scores based on Z-scores in each cell type of IPF and control group. Module 1 genes signature demonstrated the highest score in Proliferatingcells Moreover, the Module 1 signature score of Proliferatingcells was significantly higher in IPF than in control group (figure 8C).

4. Discussion

Nine upregulated markers, including NUF2, CEP55, ANLN, TTK, TK1, MYBL2, CCNA2, RRM2, and CDT1, constitute Module 1, which is primarily involved in cell division, cell cycle, and their regulation. Within this module, NUF2 occupies a central position. All nine genes exhibit the highest expression levels in "Proliferating Cells." Previous research has demonstrated the involvement of these nine genes in the pathogenesis of malignant tumors, with CCNA2 being the only gene studied in the context of IPF [13].

NUF2 is significantly associated with cell proliferation, cell division, and cell cycle arrest. In a rat model of chest irradiation-induced lung cell senescence, NUF2 is significantly upregulated and deemed a central gene. The expression levels of NUF2 in lung adenocarcinoma and lung squamous cell carcinoma are significantly higher than those in normal tissues, and patients with non-small cell lung cancer with high NUF2 expression have a poor prognosis [14]. Studies have also corroborated that NUF2 can augment the expression level of IL8, thereby stimulating the proliferation and survival of CD34+ cells in bone marrow stromal cells, leading to megakaryocyte proliferation and differentiation in primary myelofibrosis [15]. This study found that the expression level of NUF2 in BALF of IPF patients was significantly increased compared with the control group. Single-cell RNA sequencing revealed that NUF2 is predominantly highly expressed in proliferating cells.

TK1 is considered a cell cycle-dependent marker of cell proliferation, reflecting the growth rate of normal cells and various types of tumor cells, with substantial increases observed during DNA synthesis in the S phase of the cell cycle [16]. The TK1 level has been found to be related to tumor proliferation, as TK1 is proportional to the cell proliferation rate [17]. Serum TK1 activity is almost undetectable in healthy individuals but is observed in patients with malignant tumors [18]. Studies have shown that TK1, as a tumor marker, is more sensitive for tumor screening than CEA and AFP [19]. A meta-analysis revealed that TK1 overexpression is associated with a poor prognosis in lung cancer patients [15]. However, no studies have been documented on TK1 in patients with lung fibrosis.

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MYBL2, a highly conserved member of the Myb transcription factor family, plays a pivotal role in regulating cell proliferation and survival. MYBL2 is overexpressed in lung adenocarcinoma, and knockdown of MYBL2 can induce apoptosis of lung cancer cells and significantly inhibit cell proliferation, migration, and invasion [20]. Furthermore, upregulated MYBL2 gene expression is a robust predictor of poor prognosis in lung adenocarcinoma [21].

Chromatin licensing and DNA replication factor 1 (CDT1) are crucial in the cell cycle. In normal cells, precise DNA replication is achieved through licensing of each replication origin once per cell cycle [22]. Licensing is essential for initiating DNA replication at each origin and represents a highly conserved mechanism in eukaryotes [23]. Ectopic expression of CDT1 leads to excessive genome replication and genomic instability in eukaryotes [24]. The expression of CDT1 is downregulated in lung cancer tissues [25].

CEP55, a mitotic phosphoprotein, plays a vital role in cell division and the cell cycle. During the final stage of cell division, it facilitates the physical separation of two daughter cells [26]. CEP55 is significantly increased in lung cancer cells compared to normal lung tissue [27]. CEP55 overexpression is considered a potential adverse prognostic factor for various cancers, including lung adenocarcinoma [28], epithelial ovarian cancer [29], and esophageal squamous cell carcinoma [30]. CEP55 may influence tumor cell proliferation through Myc signaling, DNA repair, and the G2M checkpoint [27]. Beyond its role in cell mitosis and the cell cycle, CEP55 also plays a crucial role in the PI3K/Akt signaling pathway [31], one of the two main signaling pathways for cell survival/anti-apoptosis. A previous review by our team elaborated on the central role of the imbalance between alveolar epithelial cell apoptosis and fibroblast apoptosis in the pathogenesis of IPF [32]. CEP55 is significantly upregulated in the BALF of IPF patients compared to the control group. Upregulated CEP55 may lead to an imbalance between alveolar epithelial cell apoptosis and fibroblast apoptosis by affecting the cell cycle and the PI3K/AKT signaling pathway, thereby contributing to the pathogenesis and progression of IPF.

ANLN encodes Anillin, an actin-binding protein involved in the PI3K/PTEN signaling pathway [33]. ANLN is upregulated in various cancers [34-36]and has been shown to be a marker of poor prognosis [34]and associated with an aggressive cancer phenotype [37]. Anillin plays an important role in the invasion of breast cancer and pancreatic cancer by regulating the cell cycle [38]. There are no reports on ANLN in IPF or lung fibrosis. This study found that ANLN is significantly upregulated in the BALF of IPF patients.

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TTK, a serine, threonine, and tyrosine kinase, has multiple roles in mitosis and meiosis [38]. TTK is deregulated in several human tumors, and its upregulation has been reported in lung cancer, papillary thyroid cancer, breast cancer, gastric cancer, and other cancers, where it is associated with poor prognosis and tumor aggressiveness [39]. Studies have shown that TTK promotes epithelial-to-mesenchymal transition (EMT), a complex cellular process enabling epithelial cells to acquire a mesenchymal phenotype. EMT occurs in three biological contexts: development, wound healing and fibrosis, and tumor progression. Although occurring in three separate biological environments, EMT signaling shares some common molecular mechanisms that allow epithelial cells to dedifferentiate and acquire mesenchymal characteristics, conferring invasive and migratory capabilities to cells toward distant sites. Previous studies have not reported upregulated expression of TTK in IPF patients, but numerous studies have reported that EMT is involved in the fibrotic process of IPF [40]. In our study, TTK was significantly upregulated in the BALF of IPF patients, and TTK may participate in the pathogenesis of IPF by promoting EMT.

alveolar epithelium, with most CCNA2-expressing cells also positive for Ki-67 (a proliferation marker) and TUNEL staining in epithelium and hyaline membranes (evidence of apoptosis) [13]. It can be inferred that high CCNA2 expression represents an epithelial proliferative response, and this enhanced proliferation represents a failed compensatory response to injury, indicating that epithelial injury plays an important role in the pathogenesis of AE-IPF. Our study further confirmed that CCNA2 can be detected in the BALF of IPF patients and is significantly upregulated compared to the control group; PPI and module analysis indicated that CCNA2 is involved in Module 1; single-cell RNA sequencing revealed that CCNA2 is predominantly highly expressed in proliferating cells. Consistent with previous studies, it has been shown that CCNA2 mainly originates from proliferating cells.

RRM2 is a rate-limiting enzyme involved in DNA synthesis and damage repair, playing a crucial role in many key cellular processes such as cell proliferation, invasion, migration, and senescence [41]. The expression level of RRM2 is increased in lung cancer tissues compared to normal lung tissues, and high expression levels are associated with poor prognosis [42]. Han et al. found that overexpression of RRM2 promotes the migration and invasion of nasopharyngeal carcinoma cells by inducing EMT. RRM2 can also promote the expression of Bcl-2 [43]. Since Bcl-2 has an anti-apoptotic effect, increased RRM2 expression can upregulate Bcl-2, thereby enhancing the anti-apoptotic effect. Both EMT and fibroblast anti-apoptosis are key mechanisms involved in the pathogenesis of IPF, and RRM2 may participate in the

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pathogenesis of IPF by affecting EMT and fibroblast apoptosis.

In summary, the expression levels of the nine genes (NUF2, CEP55, ANLN, TTK, TK1, MYBL2, CCNA2, RRM2, and CDT1) are highest in "Proliferating cells." Previous studies have demonstrated their involvement in the pathogenesis of malignant tumors. Additionally, NUF2 is also associated with lung cell aging, megakaryocyte proliferation, and differentiation [15]; CEP55 may also participate in cell anti-apoptosis by affecting the PI3K/AKT signaling pathway [31]; ANLN is also involved in the PI3K/PTEN signaling pathway [33]; TTK and RRM2 promote EMT [44,45]; RRM2 can also upregulate Bcl-2 [43], leading to increased anti-apoptotic effects; MYBL2 is also involved in regulating cell proliferation and survival [20]. Only CCNA2 has been studied in the context of IPF [13]. In fact, lung cell aging, fibroblast proliferation, alveolar epithelial cell apoptosis, fibroblast anti-apoptosis, EMT, and the PI3K/PTEN signaling pathway are also important mechanisms involved in the pathogenesis of IPF. Although IPF and lung cancer appear unrelated, some scholars have elaborated on the intricate connections between them.

The main limitation of this study is the lack of sufficient experimental validation. Bulk RNA-seq was performed on BALF, which led to the identification of a proliferation-related gene module (Module 1, including NUF2, CEP55, ANLN, TTK, TK1, MYBL2, CCNA2, RRM2, and CDT1). Single-cell RNA-seq was subsequently carried out on lung tissues from a separate cohort of 4 IPF patients and 3 controls. The Module 1 genes, originally derived from BALF, were found to be predominantly expressed in proliferating cells within lung tissues. Moreover, the Module 1 signature score of these proliferating cells was significantly higher in the IPF group than in the control group. This constitutes the sole validation step in our study: markers highly expressed in te BALF of one cohort were corroborated at the transcriptional level in lung tissue samples from another cohort, though both approaches relied solely on sequencing technologies. No additional validation was performed for the proposed markers. Specifically, we did not conduct RT-qPCR to confirm the expression of Module 1 genes in BALF cell pellets or lung tissues. Furthermore, due to the unavailability of remaining lung tissue samples, we were unable to perform experimental validations such as immunohistochemistry, immunofluorescence. To address this limitation, we could obtain specimens via transbronchial lung cryobiopsy in the future. This would allow us to acquire sufficient samples not only for sequencing but also for subsequent validation experiments such as RT-qPCR, immunohistochemistry, and immunofluorescence.

5. Conclusion

In the comparison between the IPF group and the control group, bulk RNA-seq was performed on BALF samples, followed by differential expression analysis, which identified 104 DEGs.

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Among the upregulated markers, NUF2, CEP55, ANLN, TTK, TK1, MYBL2, CCNA2, RRM2, and CDT1 form a functional module primarily involved in cell division, cell cycle regulation, and associated processes. These markers have been more extensively studied in the context of malignant tumors. Single-cell sequencing of cell populations/subpopulations revealed that these nine markers are most highly expressed in proliferating cell populations. This suggests that IPF also exhibits characteristics akin to tumor cell unlimited proliferation. Although the proliferating cells in the lung tissue of IPF patients are not heterogeneous, they may share some pathophysiological mechanisms with tumors, representing a future research avenue. Some targeted antitumor drugs may also be effective for IPF, and further animal experiments are warranted to validate this hypothesis.

Conflicts of interest statement

The authors have no conflict of interest.

Date availability statement

The datasets generated and/or analyzed during the current study are available in the GEO repository (To review GEO accession GSE312052:Go to https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE312052 Enter token kjalqsoovrwzzqr into the box).

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Figure Legends

Figure 1. Technical Route Outlined

Abbreviations: BALF : bronchoalveolar lavage fluid; bulk RNA-seq: bulk RNA sequencing; TBLB: transbronchial lung biopsies; DEGs: Differentially Expressed Genes.

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Figure 2. Analysis of DEG Expression Patterns between Groups

IPF group: S1-S14; Control group: S15-S20.

Figure 3. IPF vs Control Group GO Level 4 Classification Bar Chart

The x-axis represents GO terms, divided into three major categories from left to right: biological process, cellular component, and molecular function. The y-axis represents the number of genes in the term. Red indicates upregulated genes, and green indicates downregulated genes. GO, Gene Ontology.

Figure 4 KEGG Enrichment Bubble Chart

Showing the top 20 upregulated signaling pathways with the smallest Q value (Figure A) and the top 20 downregulated signaling pathways with the smallest Q value (Figure B). The y-axis represents the signaling pathway name, and the x-axis represents the enrichment factor (RichFactor). GO, Gene Ontology; KEGG, Kyoto Gene and Genome Encyclopedia.

Figure 5: Functional Modules Obtained from MCODE Analysis in Cytoscape.

Each node represents a protein, and each line represents the interaction between proteins. Red nodes represent upregulated genes, and blue nodes represent downregulated genes.

Figure 6 Correlation between TTK and NUF2 with TLC%

TTK : TTK protein kinase; NUF2: NDC80 kinetochore complex component; TLC%: the percentage of measured to predicted value of total Lung CapacityTLC%.

Figure 7. UMAP Plot Colored by Cell Type (A), UMAP Plot Colored by Sample (B), and Bar Chart of Single-Sample Cell Composition (C)

Figure 8 Expression of Module 1 genes in Different Cell Populations(A) and Proliferating Cell Subpopulations(B), and Module 1 genes signature scores(C).

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Table 1. Clinical, Laboratory, and Examination Characteristics of Subjects

	Control Group	IPF Group		Control Group	IPF Group
Number of Subjects	6	14	Blood Routine		
Age	60.33 ± 11.48	64.00 ± 9.29	White Blood Cells	5.49 ± 1.07	**7.01 ± 1.53 ***
Gender (Male/Female)	2/4	7/7	Platelets	211.50 ± 42.86	249.14 ± 85.44
Smoking History (Smokers/ Non-smokers)	3/3	8/6	NEU%	0.48 ± 0.13	0.61 ±0.13
BALF Flow Cytometry			L%	0.39 ± 0.12	**0.26 ± 0.11 ***
CD4+/CD8+ T Cells	1.69 ± 0.98	0.93 ± 0.47 *	M%	0.09 ± 0.01	0.08 ± 0.02
Macrophages	0.58 ± 0.14	0.58 ± 0.13	E%	0.02 ± 0.02	0.04 ± 0.02
Lymphocytes	0.23 ± 0.10	0.16 ± 0.14	NEU Absolute Value	2.71 ± 1.07	**4.40 ± 1.66 ***
Neutrophils	0.18 ± 0.08	0.26 ± 0.09	L Absolute Value	2.07 ± 0.42	1.72 ± 0.42
Eosinophils	0.0027 ± 0.0012	0.0031 ± 0.0027	M Absolute Value	0.48 ± 0.08	0.53 ± 0.08
Peripheral Blood Flow Cytometry			E Absolute Value	0.20 ± 0.11	0.29 ± 0.15
T Lymphocytes	0.79 ± 0.07	0.73 ± 0.09	Serum Tumor Markers		
B Lymphocytes	0.13 ± 0.061	0.11 ± 0.03	Ferritin (ng/mL)	100.18 ± 83.92	198.08 ± 138.77
NK Cells	0.09 ± 0.04	0.15 ± 0.09	CEA (ng/ml)	3.40 ± 2.70	3.45 ± 2.53
CD4+ T Cells	0.51 ± 0.10	0.41 ± 0.10	CA199 (U/ml)	8.14 ± 3.27	54.96 ± 116.23
CD8+ T Cells	0.24 ± 0.07	0.30 ± 0.10	Cytokeratin 19 Fragment (ng/mL)	1.76 ± 0.58	**3.30 ± 0.81 ***
CD4+/CD8+ T Cells	-	1.57 ± 0.83	Squamous Cell Carcinoma Antigen (SCC) (ug/mL)	0.90 ± 0.20	0.97 ± 0.36

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Blood Gas Analysis			ProGRP (pg/L)	29.03 ± 3.39	30.21 ± 13.32
PaO2	75.92 ± 11.78	73.68 ± 13.51	Other		
AaDO2	28.2 ± 14.61	34.25 ± 22.41	Fib (g/L)	2.62 ± 0.65	3.88 ± 1.36
PaO2/FiO2	361.52 ± 56.07	338.37 ± 9.96	D-Dimer (ng/mL)	463.6 ± 242.05	540.79 ± 241.87
Lung Function			CRP (ng/dL)	-	895.92 ± 440.40
FVC% Predicted	1.03 ± 0.09	0.96 ± 0.49	ALB (g/L)	0.43 ± 0.32	2.60 ± 5.64
FEV1% Predicted	0.98 ± 0.13	0.99 ± 0.56	LDH (U/L)	11.60 ± 6.19	**26.00 ± 20.18 ***
DLCO% Predicted	0.88 ± 0.29	0.61 ± 0.19 ∗	autoantibody panel	negative	negative
TLC% Predicted	0.86 ± 0.16	0.68 ± 0.13 ∗			

Notes:
PaO2: Partial Pressure of Oxygen; AaDO2: Alveolar-Arterial Oxygen Gradient; PaO2/FiO2: Oxygenation Index; FVC: Forced Vital Capacity; FEV1: Forced Expiratory Volume in 1 Second; DLCO: Diffusing Capacity for Carbon Monoxide; TLC: Total Lung Capacity; NEU: Neutrophils; L: Lymphocytes; M: Monocytes; E: Eosinophils; Fib: Fibrinogen; D-Dimer: D-Dimer; CRP: C-Reactive Protein; ALB: Albumin; LDH: Lactate Dehydrogenase. %: Percentage; ∗: $p < 0.05$.

Table 2. The differentially expressed genes in IPF compared to the control group.

Gene	**Full name/description of gene**	**IPF _fpkm**	**Control _fpkm**	**FDR**	**FC**
	A, Upregulated				
CCL13	C-C motif chemokine ligand 13	14.032	0.333	0.000	32.33
NUF2	NDC80 kinetochore complex component	0.538	0.010	0.016	29.87
IGLL5	immunoglobulin lambda like polypeptide 5	4.886	0.195	0.009	26.72
TUBB3	tubulin beta 3 class III	3.038	0.080	0.004	25.27
MEGF6	multiple EGF like domains 6	0.249	0.005	0.003	22.48
MYBL2	MYB proto-oncogene like 2	1.651	0.073	0.001	20.67
CTSE	cathepsin E	0.469	0.015	0.009	19.48
TMIGD3	transmembrane and immunoglobulin domain containing 3	5.530	0.275	0.023	14.94
KRT6A	keratin 6A	1.887	0.312	0.030	12.41
TTK	TTK protein kinase	0.999	0.177	0.002	11.30
CEP55	centrosomal protein 55	1.130	0.093	0.016	10.11
CSGALNACT1	chondroitin sulfate N-acetylgalactosaminyltransferase 1	0.629	0.153	0.046	8.96
RRM2	ribonucleotide reductase regulatory subunit M2	4.775	1.012	0.009	7.68
ANLN	anillin actin binding protein	1.000	0.358	0.031	6.58
MERTK	MER proto-oncogene, tyrosine kinase	1.198	0.223	0.022	6.55
CDT1	chromatin licensing and DNA replication factor 1	1.355	0.173	0.042	6.16
TK1	thymidine kinase 1	4.821	0.757	0.006	6.09
PTPN7	protein tyrosine phosphatase, non-receptor type 7	2.874	0.640	0.026	4.87

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CCNA2	cyclin A2	0.771	0.135	0.009	4.75
RASSF2	Ras association domain family member 2	6.725	1.783	0.005	3.61
ANKH	ANKH inorganic pyrophosphate transport regulator	4.789	2.583	0.046	2.28
TCONS_00090454	> P01859.2 RecName: Full=Immunoglobulin heavy constant gamma 2	7.958	0.103	0.042	53.09
TCONS_00090453	> P01857.1 RecName: Full=Immunoglobulin heavy constant gamma 1	26.494	1.390	0.042	13.21
TCONS_00090461	> AAS01770.1 monoclonal IgM antibody heavy chain	9.505	0.167	0.026	31.19
	B, Downregulated				
SUSD1	sushi domain containing 1	4.516	7.787	0.044	0.50
GM2A	GM2 ganglioside activator	8.196	14.097	0.036	0.49
HLA-DRB1	major histocompatibility complex, class II, DR beta 1	176.759	280.055	0.036	0.49
HLA-DRA	major histocompatibility complex, class II, DR alpha	534.629	907.143	0.016	0.49
UNC93B1	unc-93 homolog B1	5.762	9.622	0.030	0.48
SLC15A3	solute carrier family 15 member 3	29.314	52.350	0.045	0.48
HLA-DMA	major histocompatibility complex, class II, DM alpha	75.636	132.707	0.003	0.46
HLA-DPA1	major histocompatibility complex, class II, DP alpha 1	198.577	354.827	0.003	0.45
CD37	CD37 molecule	54.187	119.858	0.045	0.45
ANKHD1	ankyrin repeat and KH domain containing 1	6.666	12.168	0.035	0.43
LYZ	lysozyme	223.384	407.753	0.001	0.42
NUPR1	nuclear protein 1, transcriptional regulator	106.514	191.955	0.003	0.42
TSPYL2	TSPY-like 2	2.517	5.203	0.006	0.39
APOC1	apolipoprotein C1	363.112	684.093	0.031	0.39
REL	REL proto-oncogene, NF-kB subunit	2.621	5.562	0.049	0.36
SLC47A1	solute carrier family 47 member 1	2.221	2.900	0.035	0.35
P4HA2	prolyl 4-hydroxylase subunit alpha 2	2.589	5.413	0.022	0.34
LY86	lymphocyte antigen 86	10.991	20.318	0.029	0.34
RBP4	retinol binding protein 4	9.249	21.043	0.034	0.34
ADTRP	androgen dependent TFPI regulating protein	8.349	20.945	0.002	0.33
ANGEL1	angel homolog 1 (Drosophila)	0.815	1.667	0.012	0.33
Septin-4	septin 4	3.606	7.417	0.022	0.33
Septin-3	septin 3	0.968	2.870	0.004	0.29
SERPING1	serpin family G member 1	21.135	57.197	0.000	0.29
CD6	CD6 molecule	0.446	1.095	0.040	0.29
RPAP2	RNA polymerase II associated protein 2	1.671	3.752	0.031	0.29
PDCD1LG2	programmed cell death 1 ligand 2	1.671	4.560	0.000	0.28
ZNF365	zinc finger protein 365	1.936	4.788	0.033	0.28
TMEM182	transmembrane protein 182	6.274	23.780	0.009	0.28
SULT1B1	sulfotransferase family 1B member 1	1.520	4.178	0.030	0.27
KPNA5	karyopherin subunit alpha 5	1.143	2.350	0.015	0.25
WDPCP	WD repeat containing planar cell polarity effector	4.059	10.557	0.003	0.24
PIGN	phosphatidylinositol glycan anchor biosynthesis class N	9.861	30.345	0.009	0.21
NEDD4	neural precursor cell expressed, developmentally down-regulated 4, E3 ubiquitin protein ligase	24.949	78.255	0.006	0.21
PNPLA3	patatin like phospholipase domain containing 3	0.626	2.315	0.017	0.20
P2RX1	purinergic receptor P2X 1	0.351	1.235	0.023	0.20
ICOS	inducible T-cell costimulator	0.509	2.098	0.039	0.19
C4orf32	chromosome 4 open reading frame 32	2.940	11.572	0.004	0.19
RBM11	RNA binding motif protein 11	0.494	2.267	0.011	0.19
CES1	carboxylesterase 1	20.467	161.992	0.000	0.18
NMNAT3	nicotinamide nucleotide adenylyltransferase 3	6.244	23.363	0.002	0.17
KIAA1328	KIAA1328	6.859	30.417	0.002	0.17
MAB21L3	mab-21 like 3	0.227	0.935	0.049	0.16
OR14J1	olfactory receptor family 14 subfamily J member 1	0.109	0.575	0.010	0.16
OR4F17	olfactory receptor family 4 subfamily F member 17	8.176	37.458	0.002	0.16
C8B	complement component 8, beta polypeptide	1.528	8.360	0.016	0.16
PCDHA4	protocadherin alpha 4	0.107	0.395	0.035	0.15
FSIP2	fibrous sheath interacting protein 2	1.668	8.237	0.000	0.15
TBC1D8B	TBC1 domain family member 8B	5.252	21.810	0.000	0.15
IDUA	iduronidase, alpha-L-	0.522	1.932	0.039	0.13
RPH3A	rabphilin 3A	0.166	1.400	0.001	0.09
SPINK9	serine peptidase inhibitor, Kazal type 9	0.281	2.695	0.046	0.07
SOGA3	SOGA family member 3	0.036	0.153	0.029	0.04
IGFN1	immunoglobulin-like and fibronectin type III domain containing 1	0.001	0.032	0.034	0.03
CTRB2	chymotrypsinogen B2	0.040	2.482	0.000	0.02
UPP2	uridine phosphorylase 2	0.001	0.345	0.042	0.02
MSMP	microseminoprotein, prostate associated	0.001	1.153	0.046	0.02

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RPA4	replication protein A4	0.001	0.250	0.030	0.02
TBC1D3I	TBC1 domain family member 3I	0.008	0.717	0.035	0.01
PXDNL	peroxidasin like	0.001	1.432	0.016	0.01
COL20A1	collagen type XX alpha 1 chain	0.001	0.147	0.012	0.01
KRTAP8-1	keratin associated protein 8-1	0.001	1.838	0.030	0.01
EMX1	empty spiracles homeobox 1	0.052	1.562	0.000	0.01
SPATA21	spermatogenesis associated 21	0.001	0.218	0.009	0.01
CTD-3222D19.2	-	0.001	0.177	0.049	0.01
CTD-2192J16.22	-	0.008	2.277	0.016	0.01
TCONS_00266883	> XP_020145385.1 gamma-secretase-activating protein isoform X5	0.904	3.243	0.040	0.24
TCONS_00306038	-	3.416	13.038	0.000	0.22
TCONS_00041674	>Q9UHZ2.1 RecName: Full=Metastasis-associated lung adenocarcinoma transcript 1	3.134	25.753	0.000	0.08
TCONS_00258771	> EAW48689.1 hCG1817353	0.718	2.790	0.031	0.20
TCONS_00209162	> EAW78716.1 hCG1813636	0.344	6.908	0.013	0.06
TCONS_00026971	-	0.001	3.165	0.042	0.02
TCONS_00294094	> EDL08604.1 mCG1030012	0.001	1.157	0.041	0.02
TCONS_00201285	-	0.001	13.950	0.045	0.02
TCONS_00283606	-	0.038	7.328	0.006	0.01
TCONS_00110939	> XP_018867249.1 PREDICTED: LOW QUALITY PROTEIN: AFG3-like protein	0.001	3.360	0.026	0.01
TCONS_00021185	-	0.001	14.817	0.031	0.01
TCONS_00140025	-	0.001	2.758	0.003	0.01

FC, fold change; FDR, adjusted p-value.

Table 3. Characteristics of 4 Samples for Single-Cell Sequencing

Sample ID	Gender	Age	Race	Smoking	Sampling Method	Sampling Site	Cell Count	Histopathological Result
IPF1	Female	63	Asia	No	TBLB	Right Upper Lobe	8099	UIP
IPF2	Male	57	Asia	No	TBLB	Right Lower Lobe	5601	UIP
IPF3	Male	68	Asia	Yes	TBLB	Right Lower Lobe	1157	UIP
IPF4	Male	67	Asia	No	TBLB	Right Upper and Middle Lobes	3683	UIP

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Sample ID	Gender	Age	Race	Smoking	Sampling Method	Sampling Site	Cell Count	Histopathological Result
control 1	Female	55	Asia	No	TBLB	Right Upper Lobe	3729	Normal
Control2	Male	68	Asia	Yes	TBLB	Right Lower Lobe	3583	Normal
Control3	Female	30	Male	No	TBLB	Right Lower Lobe	4625	Normal

Note: TBLB refers to Transbronchial Lung Biopsy. UIP refers to Usual Interstitial Pneumonia.

Table 4: Cell Type/Abbreviation/Marker Correspondence Table

Abbreviation	Cell Type	Marker
TCells	T Cells	TRBC1, TRBC2, CD2, CD3D
MPs	Monocytes/Macrophages	C1QA, MRC1, CD163, CD1C
MastCells	Mast Cells	TPSAB1, TPSB2, CPA3
PlasmaCells	Plasma Cells	JCHAIN, MZB1, CD79A, IGKC
ECs	Endothelial Cells	CDH5, CLDN5, VWF, PECAM1
CiliatedCells	Ciliated Cells	FOXJ1, PIFO, TPPP3
SecretoryCells	Secretory Cells	WFDC2, MUC5AC, MUC5B, SCGB1A1, SCGB3A1
BasalCells	Basal Cells	KRT15, KRT17, KRT5, KRT19
ProliferatingCells	Proliferating Cells	MKI67, TOP2A, STMN1
AlveolarEpithelialCells	Alveolar Epithelial Cells	SFTPB, SFTPC, ABCA3, LAMP3, CAV1, AGER, CLIC5
StromalCells	Stromal Cells	COL1A1, DCN, LUM, ACTA2, TAGLN

Note: The table lists various cell types along with their abbreviations and corresponding markers.

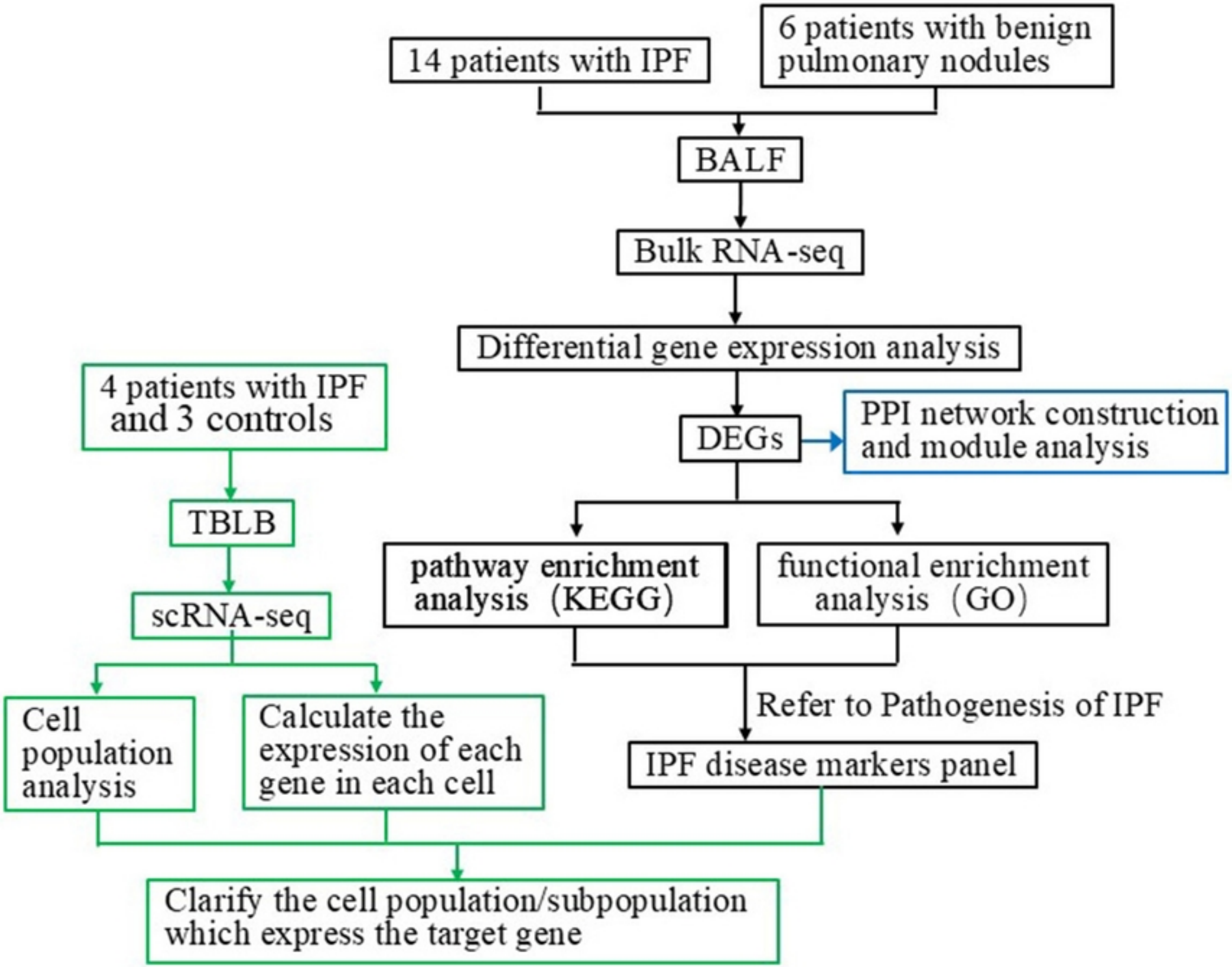
14 patients with IPF
6 patients with benign pulmonary nodules
BALF
Bulk RNA-seq
Differential gene expression analysis
DEGs
PPI network construction and module analysis
pathway enrichment analysis (KEGG)
functional enrichment analysis (GO)
Refer to Pathogenesis of IPF
IPF disease markers panel
4 patients with IPF and 3 controls
TBLB
scRNA-seq
Cell population analysis
Calculate the expression of each gene in each cell
Clarify the cell population/subpopulation which express the target gene

Figure 1

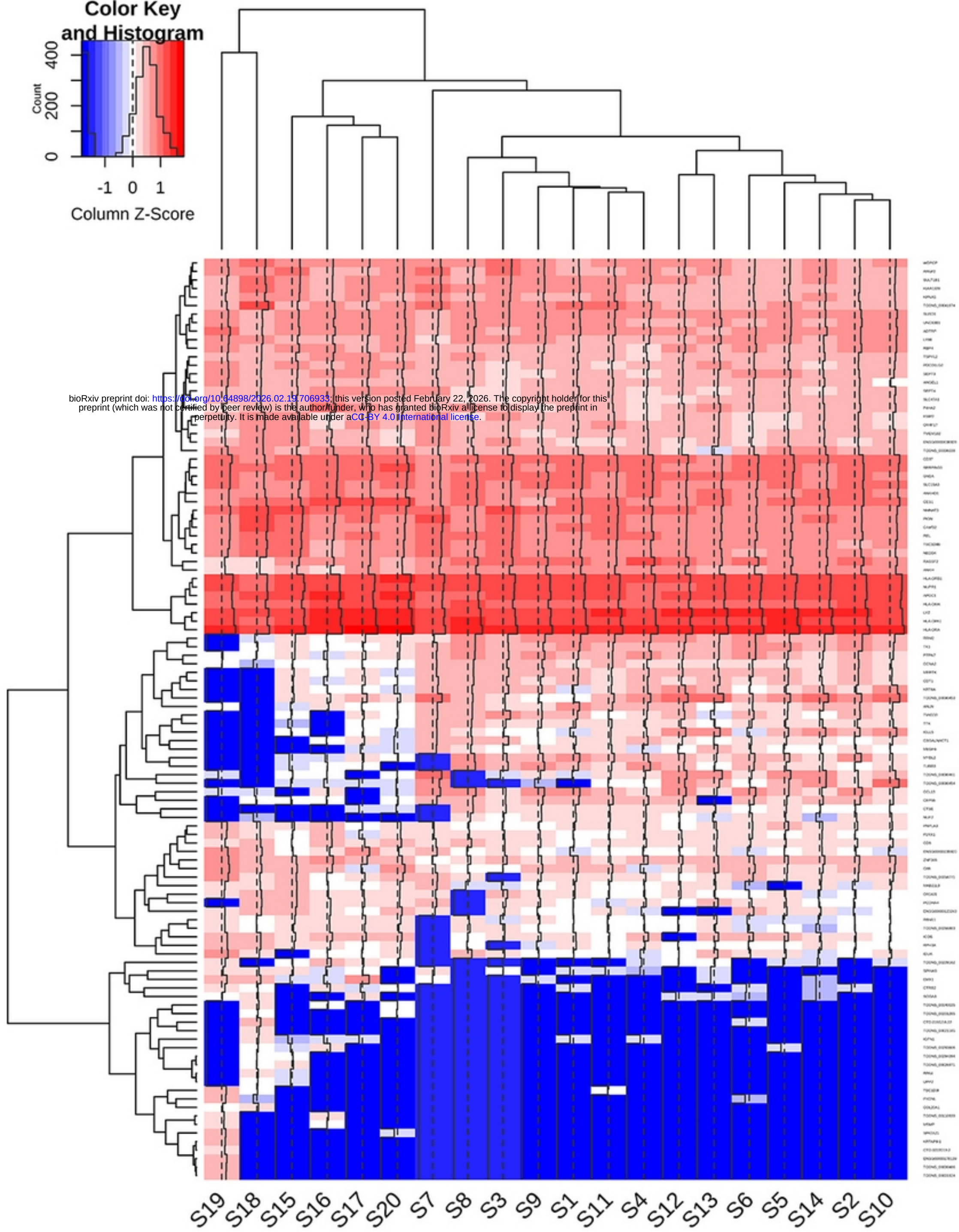

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Figure 2

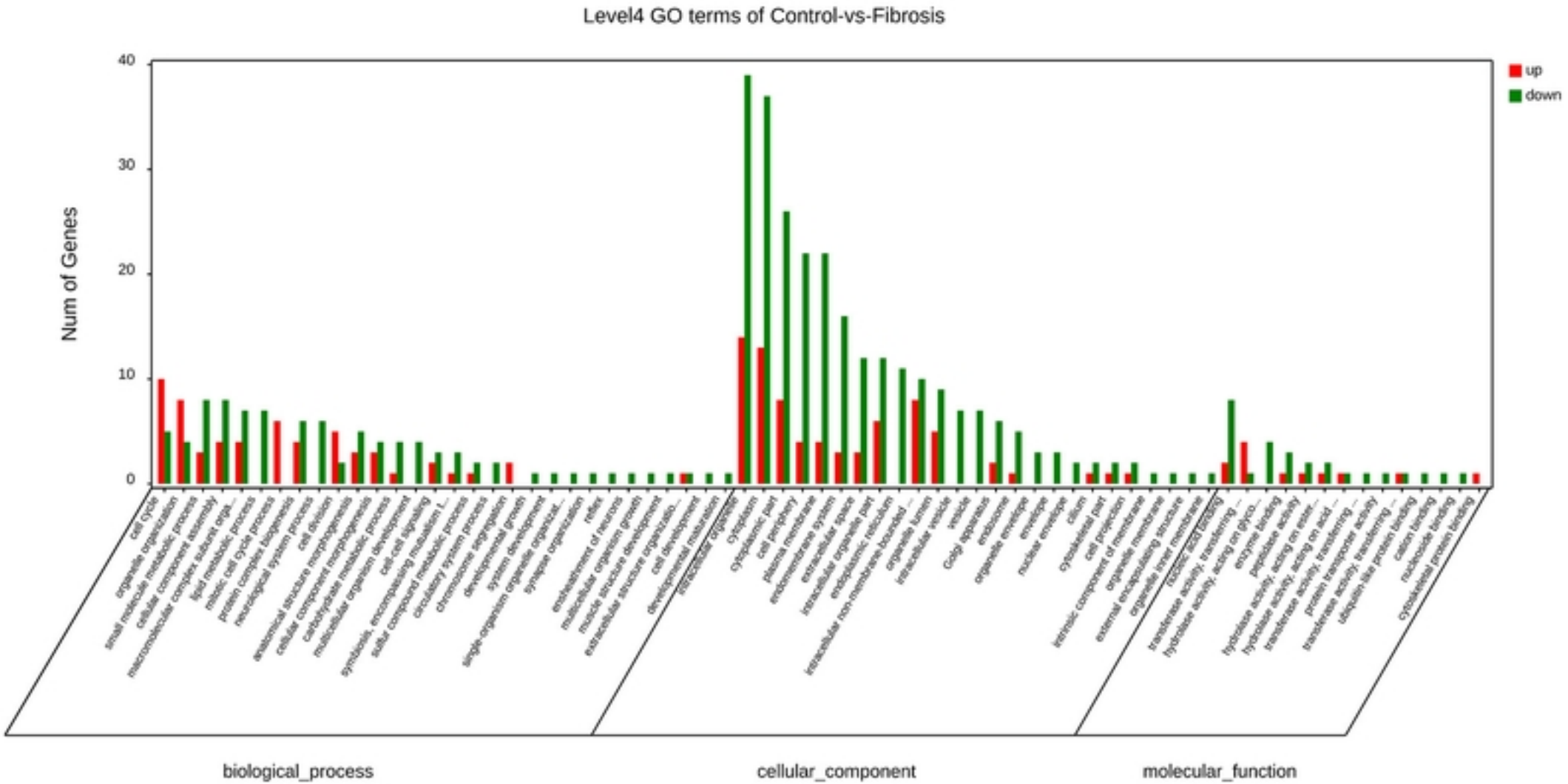

Level4 GO terms of Control-vs-Fibrosis
Num of Genes
40
30
20
10
0
up
down
cell cycle
organelle organization
small molecule metabolic process
cellular component assembly
macromolecular complex subunit orga...
lipid metabolic process
mitotic cell cycle process
protein complex biogenesis
neurological system process
cell division
anatomical structure morphogenesis
cellular component morphogenesis
carbohydrate metabolic process
multicellular organism development
cell-cell signaling
symbiosis, encompassing mutualism t...
sulfur compound metabolic process
circulatory system process
chromosome segregation
developmental growth
system development
single-organism organelle organizat...
synapse organization
reflex
establishment of neuron
multicellular organism growth
muscle structure development
extracellular structure organizatio...
cell development
developmental maturation
extracellular organelle
cytosolic part
cytoplasmic part
cell periphery
plasma membrane
endomembrane system
extracellular space
intracellular organelle part
endoplasmic reticulum
intracellular non-membrane-bounded ...
organelle lumen
intracellular vesicle
vesicle
Golgi apparatus
endosome
organelle envelope
envelope
nuclear envelope
cilium
cytoskeletal part
cell projection
intrinsic component of membrane
organelle membrane
external encapsulating structure
organelle inner membrane
nucleic acid binding
transferase activity, transferring ...
hydrolase activity, acting on glyco...
enzyme binding
peptidase activity
hydrolase activity, acting on ester...
hydrolase activity, acting on acid ...
transferase activity, transferring ...
protein transporter activity
transferase activity, transferring ...
ubiquitin-like protein binding
cation binding
nucleoside binding
cytoskeletal protein binding
biological_process
cellular_component
molecular_function

Figure 3

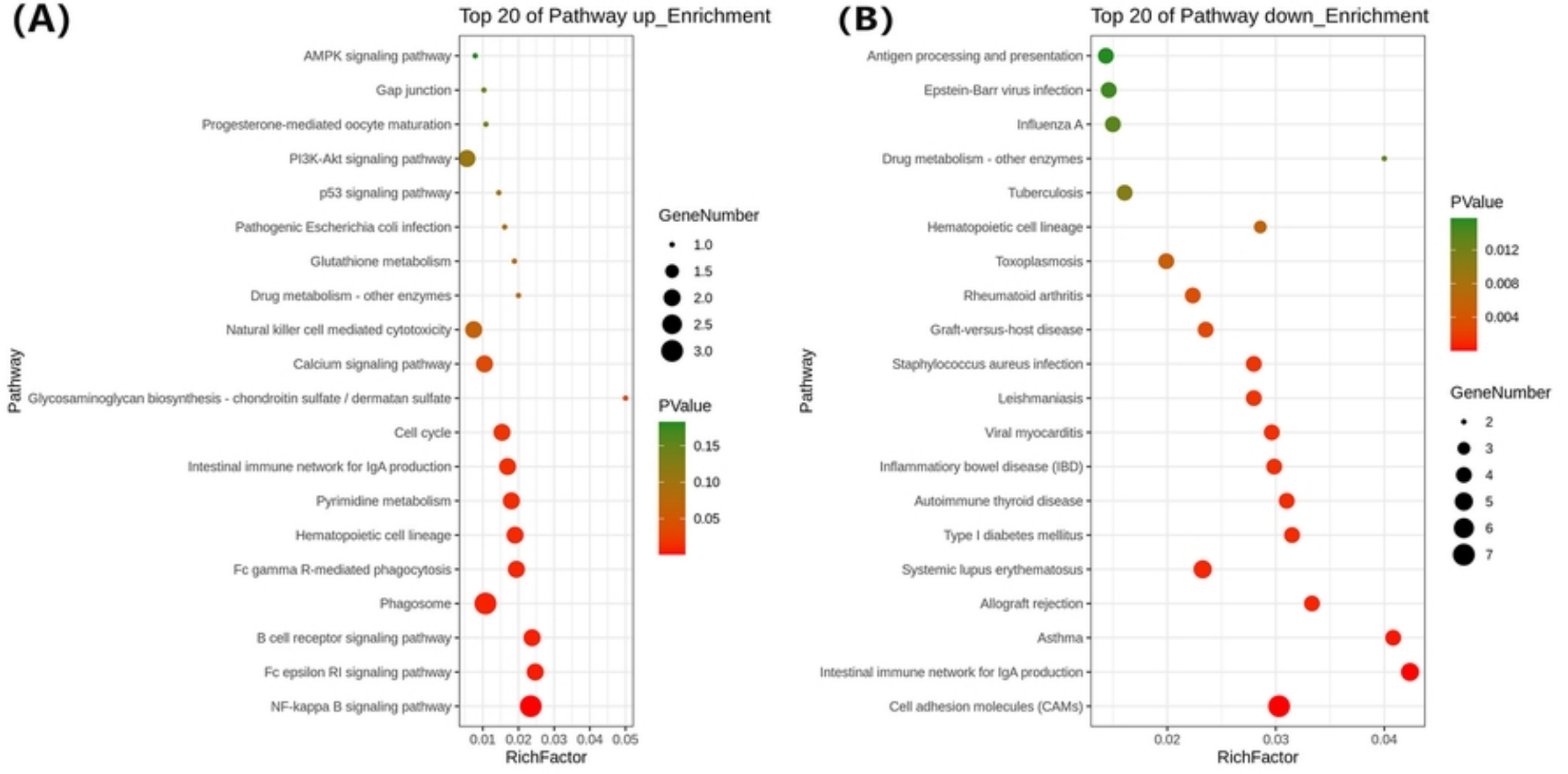

(A)
Top 20 of Pathway up_Enrichment
AMPK signaling pathway
Gap junction
Progesterone-mediated oocyte maturation
PI3K-Akt signaling pathway
p53 signaling pathway
Pathogenic Escherichia coli infection
Glutathione metabolism
Drug metabolism - other enzymes
Natural killer cell mediated cytotoxicity
Calcium signaling pathway
Glycosaminoglycan biosynthesis - chondroitin sulfate / dermatan sulfate
Cell cycle
Intestinal immune network for IgA production
Pyrimidine metabolism
Hematopoietic cell lineage
Fc gamma R-mediated phagocytosis
Phagosome
B cell receptor signaling pathway
Fc epsilon RI signaling pathway
NF-kappa B signaling pathway
Pathway
0.01 0.02 0.03 0.04 0.05
RichFactor
GeneNumber
1.0
1.5
2.0
2.5
3.0
PValue
0.15
0.10
0.05
(B)
Top 20 of Pathway down_Enrichment
Antigen processing and presentation
Epstein-Barr virus infection
Influenza A
Drug metabolism - other enzymes
Tuberculosis
Hematopoietic cell lineage
Toxoplasmosis
Rheumatoid arthritis
Graft-versus-host disease
Staphylococcus aureus infection
Leishmaniasis
Viral myocarditis
Inflammatiory bowel disease (IBD)
Autoimmune thyroid disease
Type I diabetes mellitus
Systemic lupus erythematosus
Allograft rejection
Asthma
Intestinal immune network for IgA production
Cell adhesion molecules (CAMs)
Pathway
0.02 0.03 0.04
RichFactor
PValue
0.012
0.008
0.004
GeneNumber
2
3
4
5
6
7

Figure 4

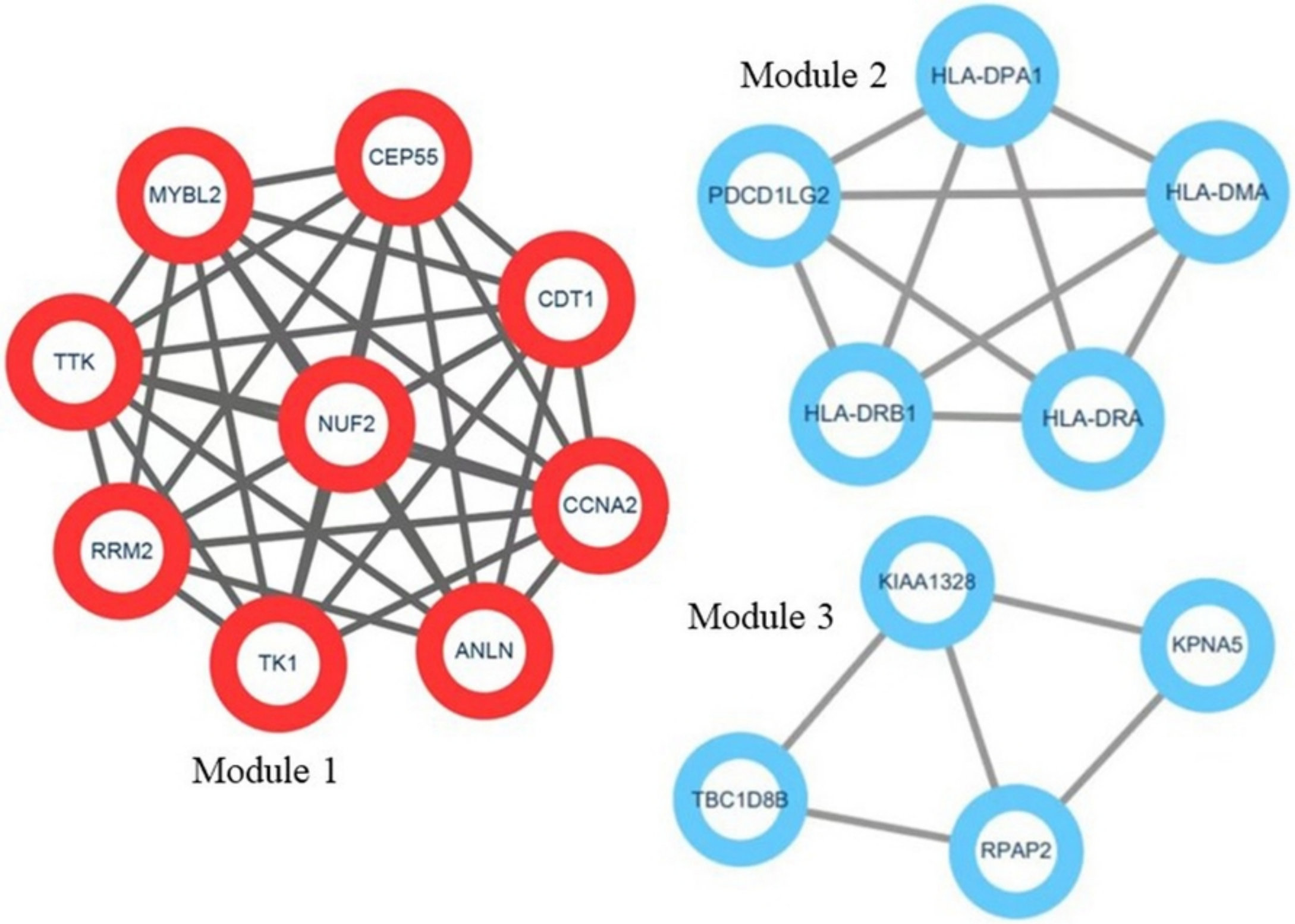
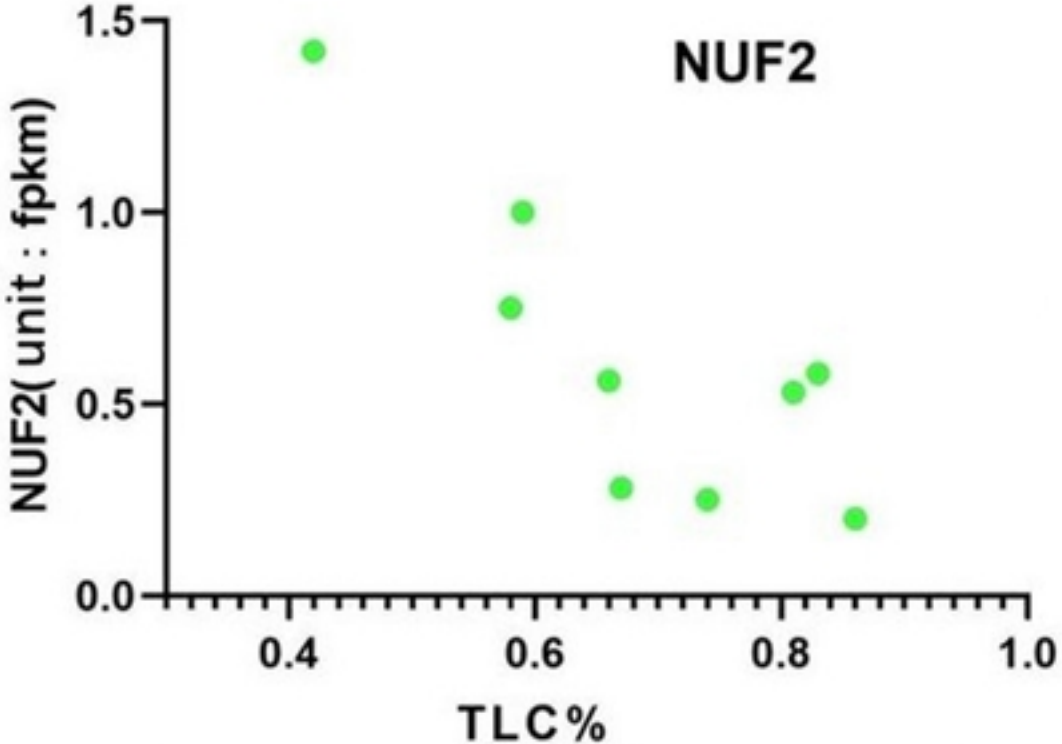
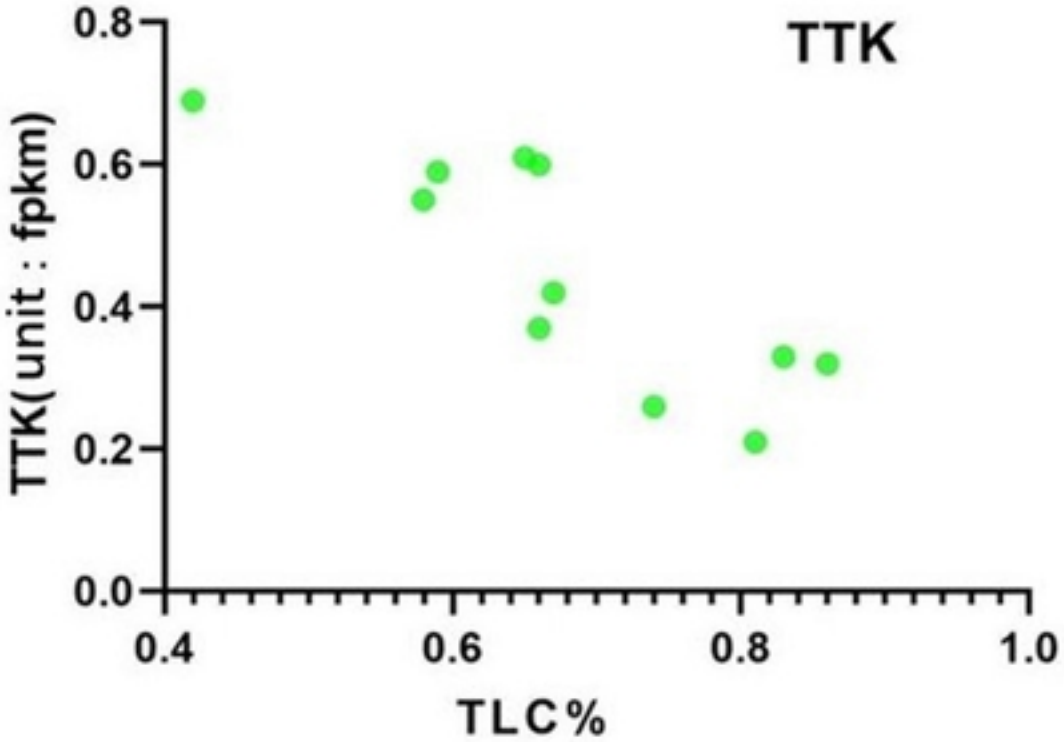

Module 1
MYBL2
CEP55
CDT1
TTK
NUF2
CCNA2
RRM2
TK1
ANLN
Module 2
HLA-DPA1
PDCD1LG2
HLA-DMA
HLA-DRB1
HLA-DRA
Module 3
KIAA1328
KPNA5
TBC1D8B
RPAP2

Figure 5



A: NUF2 was negatively correlated with TLC%



B: TTK was negatively correlated with TLC%

Figure 6

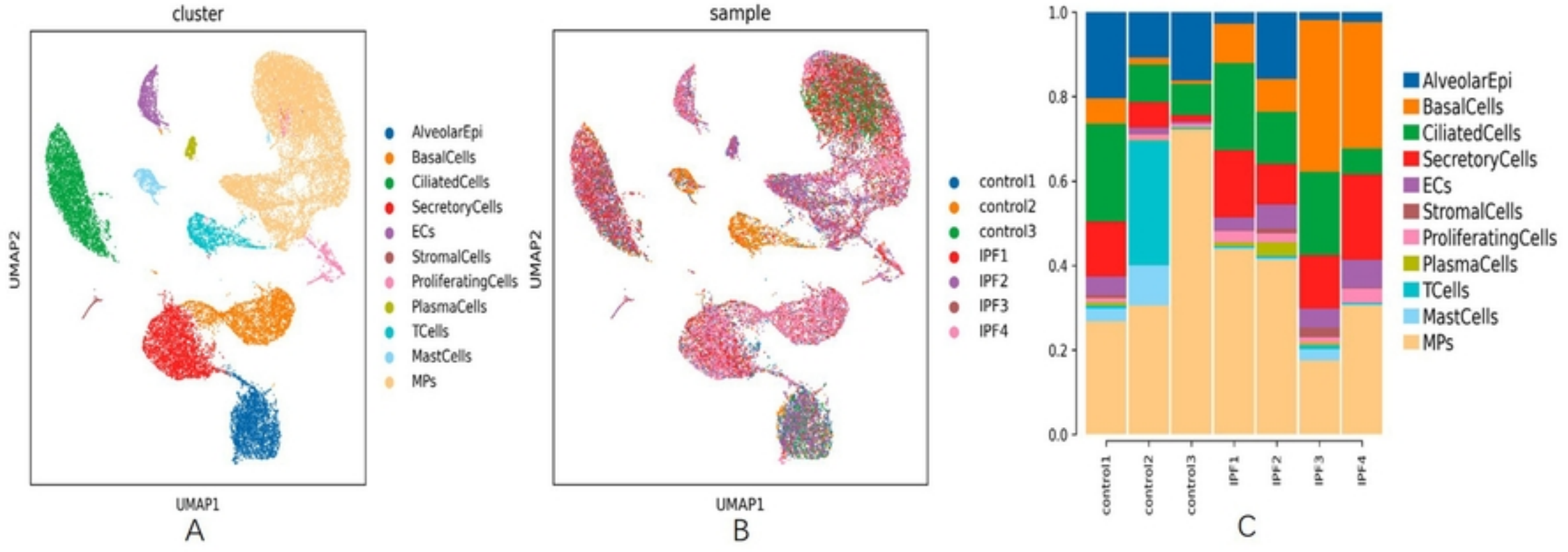

cluster
UMAP2
UMAP1
A
AlveolarEpi
BasalCells
CiliatedCells
SecretoryCells
ECs
StromalCells
ProliferatingCells
PlasmaCells
TCells
MastCells
MPs
sample
UMAP2
UMAP1
B
control1
control2
control3
IPF1
IPF2
IPF3
IPF4
1.0
0.8
0.6
0.4
0.2
0.0
control1
control2
control3
IPF1
IPF2
IPF3
IPF4
C
AlveolarEpi
BasalCells
CiliatedCells
SecretoryCells
ECs
StromalCells
ProliferatingCells
PlasmaCells
TCells
MastCells
MPs

Figure 7

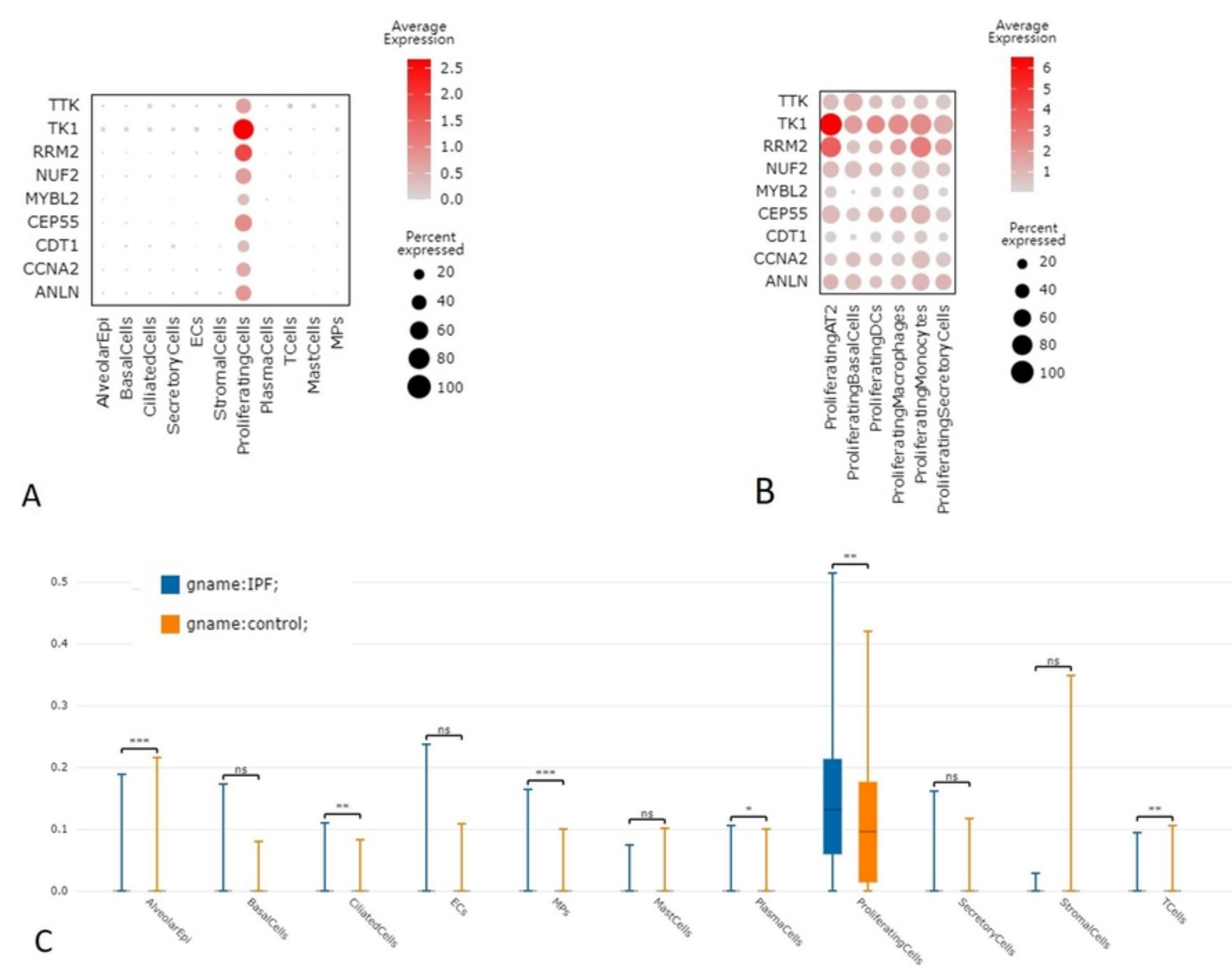

Average Expression
2.5
2.0
1.5
1.0
0.5
0.0
Percent expressed
20
40
60
80
100
TTK
TK1
RRM2
NUF2
MYBL2
CEP55
CDT1
CCNA2
ANLN
AlveolarEpi
BasalCells
CiliatedCells
SecretoryCells
ECs
StromalCells
ProliferatingCells
PlasmaCells
TCells
MastCells
MPs
A
Average Expression
6
5
4
3
2
1
Percent expressed
20
40
60
80
100
ProliferatingAT2
ProliferatingBasalCells
ProliferatingDCs
ProliferatingMacrophages
ProliferatingMonocytes
ProliferatingSecretoryCells
B
gname:IPF;
gname:control;
0.5
0.4
0.3
0.2
0.1
0.0
***
ns
**
ns
***
ns
*
**
ns
ns
**
AlveolarEpi
BasalCells
CiliatedCells
ECs
MPs
MastCells
PlasmaCells
ProliferatingCells
SecretoryCells
StromalCells
TCells
C

Figure 8